Natural Language Understanding Assignment Report Sports vs Politics Text Classifier

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Abstract

This report presents a comprehensive analysis of text classification for distinguishing between sports and politics news articles. We implement and compare three machine learning techniques: Naive Bayes with Bag of Words, K-Nearest Neighbors with TF-IDF, and Logistic Regression with TF-IDF. The study covers the complete pipeline from data collection to model evaluation, providing insights into feature representation methods and their impact on classification performance. Our results show that Naive Bayes with Bag of Words achieves the best accuracy of 80%, while highlighting the challenges of working with limited datasets.

1 Introduction

Text classification is a fundamental task in Natural Language Processing that involves automatically categorizing text documents into predefined classes. In this project, we focus on binary classification to distinguish between sports and politics news articles. This task has practical applications in news aggregation, content filtering, and automated categorization systems.

The primary objectives of this project are:

- Collect and prepare a dataset of sports and politics articles
- Implement three different machine learning classifiers from scratch
- Compare multiple feature representation techniques
- Evaluate and analyze the performance of different approaches
- Identify limitations and potential improvements

2 Data Collection and Dataset Description

2.1 Data Collection Process

For this project, we manually created a realistic dataset of news articles. The data collection process involved:

1. **Manual Creation:** We wrote 50 original news articles (25 sports, 25 politics) to ensure quality and avoid copyright issues.
2. **Domain Coverage:** Sports articles cover various sports like football, basketball, tennis, cricket, and athletics. Politics articles cover elections, legislation, governance, and international relations.
3. **Balanced Dataset:** Equal representation of both classes to prevent bias.
4. **Realistic Language:** Articles use vocabulary and sentence structures typical of real news.

2.2 Dataset Statistics

Category	Number of Articles	Percentage
Sports	25	50%
Politics	25	50%
Total	**50**	**100%**

Table 1: Dataset composition showing balanced class distribution

2.3 Dataset Characteristics

The dataset exhibits the following characteristics:

- **Vocabulary Size:** Approximately 150 unique words across all documents
- **Average Article Length:** 12-15 words per article
- **Class-Specific Keywords:**
  - Sports: team, player, match, championship, game, goal, victory, tournament
  - Politics: government, parliament, election, minister, legislation, policy, senator
- **Data Split:** 80% training (40 articles), 20% testing (10 articles)

3 Methodology

3.1 Feature Representation Techniques

We implemented three feature extraction methods:

3.1.1 Bag of Words (BoW)

Bag of Words represents text as a frequency count of words, ignoring grammar and word order.

Formula:

$$\text{BoW}(w) = \text{count}(w \text{ in document}) \tag{1}$$

Example: "The team won the game" becomes {the: 2, team: 1, won: 1, game: 1}
Advantages:

- Simple to implement
- Works well for short documents
- Captures word importance through frequency

Disadvantages:

- Loses word order information
- Common words get high weights
- Sparse representation for large vocabularies

3.1.2 TF-IDF (Term Frequency-Inverse Document Frequency)

TF-IDF weighs terms by their frequency in a document against their frequency across all documents.

Formula:

$$\text{TF-IDF}(w, d) = \text{TF}(w, d) \times \text{IDF}(w) \tag{2}$$

where:

$$\text{TF}(w, d) = \frac{\text{count}(w \text{ in } d)}{|\text{words in } d|} \tag{3}$$

$$\text{IDF}(w) = \log\left(\frac{\text{total documents}}{1 + \text{documents containing } w}\right) \tag{4}$$

Advantages:

- Reduces weight of common words
- Highlights discriminative terms
- Better for information retrieval

Disadvantages:

- More computationally expensive
- Requires all documents for IDF calculation
- Can overweight rare terms

3.2 Classification Algorithms

3.2.1 Naive Bayes Classifier

Naive Bayes uses Bayes' theorem with the "naive" assumption that all features are independent.

Formula:

$$P(c|w_1, w_2, ..., w_n) \propto P(c) \prod_{i=1}^{n} P(w_i|c) \tag{5}$$

where:

- $P(c)$ is the prior probability of class c
- $P(w_i|c)$ is the likelihood of word w_i given class c

We use **Laplace Smoothing** to handle unseen words:

$$P(w|c) = \frac{\text{count}(w, c) + 1}{\text{total words in } c + |\text{vocabulary}|} \tag{6}$$

3.2.2 K-Nearest Neighbors (KNN)

KNN classifies documents by finding the k most similar training documents and performing majority voting.

Similarity Metric - Cosine Similarity:

$$\text{sim}(d_1, d_2) = \frac{\sum_i v_1[i] \times v_2[i]}{\sqrt{\sum_i v_1[i]^2} \times \sqrt{\sum_i v_2[i]^2}} \tag{7}$$

We used $k = 5$ neighbors for our experiments.

3.2.3 Logistic Regression

Logistic Regression learns weights for each feature using gradient descent.

Prediction Formula:

$$P(y = 1|x) = \sigma(b + \sum_i w_i x_i) \tag{8}$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.

Training: Gradient descent with learning rate = 0.1 and 50 epochs.

4 Experimental Results

4.1 Performance Metrics

We evaluate our classifiers using four metrics:

- **Accuracy:** $\frac{\text{correct predictions}}{\text{total predictions}}$
- **Precision:** $\frac{\text{true positives}}{\text{true positives}+\text{false positives}}$
- **Recall:** $\frac{\text{true positives}}{\text{true positives}+\text{false negatives}}$
- **F1-Score:** $\frac{2\times\text{precision}\times\text{recall}}{\text{precision}+\text{recall}}$

4.2 Comparative Results

Method	Features	Accuracy	F1-Score
Naive Bayes	Bag of Words	**80.00%**	**88.89%**
K-Nearest Neighbors	TF-IDF	50.00%	66.67%
Logistic Regression	TF-IDF	30.00%	46.15%

Table 2: Overall performance comparison of three classifiers

4.3 Detailed Results by Class

4.3.1 Experiment 1: Naive Bayes with Bag of Words

Class	Precision	Recall	F1-Score
Politics	100.00%	80.00%	88.89%
Sports	-	-	-

Table 3: Naive Bayes performance metrics

Analysis: Naive Bayes achieved the best performance with 80% accuracy. The perfect precision indicates no false positives for politics classification. The probabilistic approach with Laplace smoothing handles unseen words effectively.

4.3.2 Experiment 2: KNN with TF-IDF

Class	Precision	Recall	F1-Score
Politics	100.00%	50.00%	66.67%
Sports	-	-	-

Table 4: K-Nearest Neighbors performance metrics

Analysis: KNN showed moderate performance with 50% accuracy. The distance-based approach struggles with small datasets where finding similar neighbors becomes challenging.

4.3.3 Experiment 3: Logistic Regression with TF-IDF

Analysis: Logistic Regression achieved the lowest accuracy at 30%. The model likely underfit due to limited training data (40 samples) and needs more examples to learn meaningful feature weights.

5 Discussion

5.1 Key Findings

1. **Best Performer:** Naive Bayes with Bag of Words (80% accuracy)

Class	Precision	Recall	F1-Score
Politics	100.00%	30.00%	46.15%
Sports	-	-	-

Table 5: Logistic Regression performance metrics

- Works well with small datasets
- Simple probabilistic approach is effective
- BoW captures sufficient information for this task

2. **Feature Representation Impact:**
   - BoW outperformed TF-IDF for this dataset
   - TF-IDF may be too sophisticated for 50 documents
   - Simple word frequencies are discriminative enough
3. **Model Complexity vs Data Size:**
   - More complex models (Logistic Regression) need more data
   - Simpler models (Naive Bayes) generalize better with limited data
   - KNN performance limited by sparse similarity space

5.2 Why Naive Bayes Performed Best

- **Probabilistic Foundation:** Makes reasonable assumptions about word distributions
- **Laplace Smoothing:** Handles unseen words gracefully
- **Few Parameters:** Less prone to overfitting
- **Works with Small Data:** Effective even with 40 training samples

6 System Limitations

6.1 Dataset Limitations

1. **Small Dataset Size:** Only 50 articles limits model learning
2. **Simplified Language:** Short sentences may not reflect real news complexity
3. **Limited Vocabulary:** 150 unique words is far less than real-world scenarios
4. **No Cross-Domain Testing:** All test data is from the same distribution
5. **Binary Classification:** Real news has many more categories

6.2 Feature Representation Limitations

1. **No Context Understanding:** BoW and TF-IDF ignore word order
2. **No Semantic Meaning:** Cannot understand synonyms or related concepts
3. **Sparse Vectors:** Most features are zero in high-dimensional space
4. **No Multi-word Phrases:** Cannot capture "prime minister" as a single unit

6.3 Model Limitations

1. **Naive Bayes:**
   - Independence assumption is unrealistic
   - Cannot model word interactions
   - Sensitive to feature distribution
2. **KNN:**
   - Computationally expensive for large datasets
   - Sensitive to k value selection
   - Poor performance in high dimensions
3. **Logistic Regression:**
   - Needs more training data
   - Linear decision boundary may be insufficient
   - Hyperparameter tuning required

6.4 Implementation Limitations

1. **Simple Tokenization:** Only splits on whitespace
2. **No Preprocessing:** No stemming, lemmatization, or stopword removal
3. **No Cross-Validation:** Single train-test split may not be representative
4. **Fixed Train-Test Split:** Not shuffled, may introduce bias

7 Future Improvements

7.1 Dataset Enhancements

- Collect larger dataset (500+ articles)
- Include more news categories
- Add real-world news articles
- Implement cross-validation
- Balance data across multiple sources

7.2 Feature Engineering

- Implement n-grams (bigrams, trigrams)
- Add part-of-speech tagging
- Include named entity recognition
- Use word embeddings (Word2Vec, GloVe)
- Implement attention mechanisms

7.3 Model Improvements

- Try ensemble methods (Random Forest, Gradient Boosting)
- Implement neural networks (LSTM, BERT)
- Use pre-trained language models
- Perform hyperparameter optimization
- Add regularization techniques

8 Conclusion

This project successfully implemented and compared three machine learning approaches for sports vs politics text classification. Our key findings are:

1. Naive Bayes with Bag of Words achieved the best performance (80% accuracy)
2. Simple feature representations can be effective for focused domains
3. Model complexity should match dataset size
4. Small datasets favor simpler, probabilistic approaches

The project demonstrates that even with limited data, classical machine learning techniques can achieve reasonable classification performance. However, the identified limitations highlight areas for improvement, particularly in dataset size, feature representation, and model sophistication.

For production systems, we recommend:

- Collecting significantly more training data
- Using more advanced features like n-grams and embeddings
- Implementing ensemble methods or deep learning approaches
- Adding proper validation and testing procedures

9 References

1. Manning, C. D., & Schütze, H. (1999). Foundations of Statistical Natural Language Processing. MIT Press.

2. Jurafsky, D., & Martin, J. H. (2020). Speech and Language Processing (3rd ed.).

3. Ramos, J. (2003). Using TF-IDF to Determine Word Relevance in Document Queries.

4. Zhang, H. (2004). The Optimality of Naive Bayes. AA, 1(2), 3.

5. Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning. Springer.